DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
Supplementary Semester Examination - Summer 2023

Course: B. Tech. **Branch: Computer Engineering** **Semester: VII**
Subject Code & Name: BTCOC601 – Compiler Design
Max Marks: 60 **Date: 13/ 07 / 2023** **Duration: 3 Hrs.**

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		
A)	Explain the various phases of a compiler and their roles in the compilation process. Illustrate the flow of information between these phases.	Remember	**6**
B)	Discuss the lexical analysis phase of a compiler. Explain the role of the lexical analyzer and the construction of tokens from the input source code.	Remember Understand	**6**
C)	What is syntax analysis? Describe the role of a parser in this phase. Differentiate between top-down and bottom-up parsing techniques, providing examples for each.	Application Remember	**6**
Q.2	**Solve Any Two of the following.**		
A)	Explain the concept of a symbol table in compiler design. Discuss its importance and the information it stores during the compilation process.	Remember Understand	**6**
B)	Describe the different types of errors that can occur during compilation. Explain the difference between lexical, syntax, and semantic errors. Provide examples for each.	Remember Understand	**6**
C)	Explain constructing syntax trees for simple expressions involving only binary operators + and -. State the use of *Leaf* and *Node* in this syntax tree.	Understand Apply	
Q. 3	**Solve Any Two of the following.**		
A)	Discuss the role of semantic analysis in a compiler. Explain how type checking and type inference are performed during this phase. Illustrate with suitable examples.	Remember Understand	**6**
B)	Explain the intermediate code generation phase of a compiler. Describe the different intermediate representations used and their advantages and disadvantages.	Knowledge Understand	**6**
C)	What is optimization in the context of a compiler? Discuss the importance of optimization and provide examples of common optimization techniques used in compilers.	Knowledge Understand	**6**
Q.4	**Solve Any Two of the following.**		
A)	Describe the code generation phase of a compiler. Explain the different code generation techniques, such as the use of registers and stack allocation.	Understand Application	**6**
B)	What are the different types of code optimization techniques? Explain loop optimization and its benefits in detail. Provide examples to illustrate your explanation	Application Understand	**6**
C)	Construct a Predictive parsing table for the Grammar **E ->E+T | T, T->T*F | F, F->(E) | id.**	Understand Apply	**6**

.Q. 5	**Solve Any Two of the following.**		
A)	Explain the concept of runtime environments in a compiler. Discuss the stack allocation and heap allocation strategies used to manage memory during program execution.	Analyze Understand	**6**
B)	Discuss the concept of code generation for control flow statements in a compiler.	Remember Understand	**6**
C)	Explain how the assignment statement ***"position = initial + rate * 60"*** is grouped into the lexemes and mapped into the tokens passed on the syntax analyzer.	Understand Apply	**6**
	***** End *****		